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Sustainable food traceability: a comprehensive model for enhancing safety and resilience in the supply chain in the Oman food industry

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Abstract

The study aims to provide a comprehensive, sustainable, and technologically advanced traceability model that enhances the safety and resilience of the food supply chain in Oman. It also seeks to create a framework for enhancing safety and resilience in the supply chain. This framework will influence other regional supply chains to embrace food traceability and contribute towards global food safety and sustainability efforts. This study followed a positivist research philosophy and used the quantitative research approach. A structured questionnaire was administered online to collect data from 385 respondents. To improve the reliability of the research instrument, a pilot test was conducted. Respondents were selected using a snowball technique. The results were tested using structural equation modelling. The results showed that compliance can improve food safety performance through data accuracy and timeliness in food supply chains in Oman. It was also found that technology traceability systems can improve food safety performance through collaborative supply chain relationships in food supply chains in Oman. Additionally, the research showed that technology traceability systems, through cooperative supply chains, enhance sustainable practices. The integration of technology traceability systems and regulatory compliance are significant elements that foster cooperation and bolster food safety and sustainability. The study holds significant implications for various stakeholders within Oman's food industry and broader applications in similar contexts. With broader implications for the global food sector, the study offers a comprehensive framework for improving Oman's food safety and supply chain resilience. Adopting such models can result in notable gains in economic efficiency, sustainability, consumer trust, and regulatory compliance. Oman can fortify its food business and serve as a model for other nations by utilising cutting-edge technologies and encouraging cooperation among stakeholders.

Keywords Sustainable food traceability, Food safety, Supply chain resilience, Oman food industry, Technology traceability systems, Regulatory compliance and collaboration, Sustainability practices

Introduction

Supply chains (SCs) have evolved to become more dynamic, exposing them to more risks, turbulence, and disruptions. About 65% of companies report a disruption annually, while 13% of those in the year 2019 claimed that they lost millions of dollars because of these disruptions [47]. Supply chain management (SCM) is the process of configuring and operating production and logistics networks effectively and efficiently [42]. Managing the

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intra- and inter-organisational supply, transformation, and delivery processes involves numerous challenges, as it involves stakeholders, such as consumers, producers, distributors, and processors. Furthermore, in the wake of these challenges, SCM has evolved from its traditional economic focus to more of a social and environmental focus. Concepts like sustainability have compelled the field of SCM to reconsider downstream and upstream flows of goods [4, 6].

The upstream flow of products and services to suppliers is examined from the perspective of manufacturing, product recovery, and reverse logistics, while the downstream flow of goods to consumers traditionally addresses responsibility and ethical issues [1, 45]. Companies should be well-informed and strategic when balancing between corporate social responsibility and the firm's financial performance. Other studies add that the global food supply chain (FSC) faces food safety and sustainability challenges that require rapid traceability systems to help SCM address the economic, social, and environmental aspects of its operations [32, 59]. Furthermore, sustainability in SCs has been a frequent debate, raising issues of waste reduction and environmental protection as well as a promising future for SCM [59]. A key objective of the global FSC is to transfer different types of food from producers, and farmers to end consumers through an intermediate channel consisting of wholesalers, traders, processors, and regulators [39, 54].

It has also been postulated that global food supply chains should promote quality and food safety and minimise waste [50]. Therefore, sustainable food traceability is a proven model for promoting safety and resilience. Sustainable food traceability focuses on the journey of food products from their origin to the consumer in a way that is environmentally, socially, and economically responsible. This requires various technologies and systems to monitor and record food movements across the SC. Agility is necessary for achieving FSC resilience because it prepares SCs for changing responses, decisions, and unusual patterns in the business environment [55]. Agility focuses on an organisation's ability to adapt quickly and effectively to changes in the environment. Furthermore, FSCs can become more resilient if they can respond quickly and effectively to unforeseen food demand and supply mechanisms [50, 54]. They can also become more resilient due to efficiency levels, innovation efforts based on technologies and Industry 4.0, and resource planning capabilities [45]. This research article discusses the importance of sustainable food traceability practices and their impact on promoting supply chain safety and resilience by discussing the role of technological traceability systems, regulatory compliance systems, and supply chain collaboration.

Background to the study

In an increasingly interconnected and globalised world, the security and resilience of food supply chains are of paramount importance. Food traceability, the ability to track food products from production to consumption, has become a critical component for ensuring food safety, improving supply chain efficiency, and increasing consumer confidence. The need for robust traceability systems is highlighted by numerous food safety incidents and growing demands for transparency from consumers and regulators. Traceability is an important phenomenon in both developed and underdeveloped countries. It has been claimed that consumers are a crucial part of any SC as they are the end users of products or services [5]. Therefore, the entire SC should serve the end user of the product or service. According to the existing literature, food traceability is the tracking of a product and its history along an FSC from the production phase to its transportation, storage, processing, distribution, and sale [59]. In addition, it involves identifying the extracted data from previous SCs and providing this information in the next step if necessary [50, 54]. As the global FSC faces transparency and accountability issues, customers are also paying more attention to the quality and safety of the food they consume. Therefore, they force global FSC actors to provide information transparency about high-quality food products that ensure a high level of safety [5].

Food traceability systems add resilience and sustainability to the global food supply chain by minimising information asymmetry and allowing companies to implement geographical safety governance structures. Furthermore, traceability systems have been crucial in increasing consumer faith regarding food safety in sustainable agricultural development [5]. Traceability systems are crucial in building trust and providing peace of mind, increasing confidence in food systems. Conversely, growers and other producers need these sustainable food traceability systems for cost-effective quality management, which helps in continuous improvement and reducing the impact of safety hazards [5, 54]. Traceability data are crucial in the efficiency of products' recall processes and in determining liability settlements between consumers and FSC players.

Implementing sustainable traceability systems and leveraging their benefits is one of the biggest challenges for the resilience and sustainability of SCMs, as they depend on consumer awareness of food safety. According to previous studies, numerous traceability technology tools improve food quality and safety in global FSCs [24, 61]. For example, radio frequency identification (RFID) and barcodes link food to the production team, packers, and equipment used for transportation. Furthermore, these food traceability technology systems are crucial

to minimise the time and effort spent on recalls, which subsequently reduces operational costs, improves FSC sustainability performance, and increases FSC resilience [30]. In addition, environmental monitoring technologies, smart packaging indicators such as gas indicators, biosensors, freshness indicators, time or temperature indicators, and wireless sensor networks (WSN) have improved SC transparency and offered alternative analytical techniques [61].

Historic food safety events have led to increased attention to consumer health and safety. For example, the 2008 Salmonella outbreak at the American Peanut Corporation caused numerous deaths and approximately 637 illnesses. Chinese melamine contamination of infant milk caused six deaths and over 294,000 illnesses in 2008 [61]. These are also common cases that require better traceability in FSCs to improve sustainability and safety levels. Studies show that these historical cases have forced FSC companies around the world to focus more on social sustainability issues such as consumer health and safety [48]. Social sustainability focuses on the practice of maintaining social well-being and justice within a community both in the present and for future generations. In developed countries such as the United States, the Food and Drug Administration (FDA) has standards in place that direct food establishments and processors to identify the sources of food, feed, and ingredients used throughout the entire production process [56]. Likewise, the United Kingdom, Australia, Canada, Japan, and China have introduced regulatory standards that monitor food traceability systems to ensure food safety [35, 48].

Unfortunately, despite all these promises and advances in traceability systems for the food supply chain, achieving sustainability in traceability to promote safety and SC resilience is still a challenging task. The global FSC and food sector still face risks and threats in harmonising information flow and product flow [56]. Furthermore, FSC traceability depends heavily on regulatory compliance capabilities, data accuracy, technology integration, and collaborative relationships in the supply chain [24, 33]. FSC stakeholders are still grappling with these challenges to make the global FSC safer and more sustainable.

Problem statement

Food traceability systems across the world still face numerous challenges that limit their capabilities to uphold food safety and promote sustainability. For instance, researchers assert that agri-food SCs continue to face challenges of eradicating greenhouse gas emissions, biodiversity protection, and promotion of fair labour practices [7, 29]. Additionally, the global FSC still reports increased food waste, as products are lost, damaged, and contaminated [29, 35]. Different governments

and global FSC stakeholders have acknowledged the benefits of creating a more comprehensive sustainable food traceability model enhancing safety and resilience across the SC, as food traceability is currently insufficient and uncoordinated [36]. It has also been reported that these ongoing food traceability challenges occur because global FSC traceability technology systems are not addressed [24]. Despite the advancement in food traceability systems technologies and numerous global and local regulatory framework efforts, there are existing challenges to making these systems and elements more holistic and resilient to achieve increased safety and sustainability [31].

The Sultanate of Oman, with its unique geographical and economic landscape, faces unique challenges and opportunities in its food supply chain. As a country that relies heavily on food imports due to its arid climate and limited agricultural capacity, ensuring the safety and integrity of its food supply is both a critical and complex task. Oman's food industry is characterised by a diverse range of stakeholders, including international suppliers, local distributors, and retailers, all of whom play a crucial role in maintaining the quality and safety of food [36]. The existing food traceability systems in Oman lack the necessary integration which leads to difficulties in effectively tracking and tracing the origin, processing, and distribution of food products. This inadequacy poses risks in quickly identifying and resolving safety issues in the food SCs. This study aims to develop a comprehensive model for sustainable food traceability tailored to the specific needs of the Omani food industry. Ensuring the security and resilience of Oman's food supply chain is a complex but essential task. Leveraging advanced technologies such as blockchain, the Internet of Things (IoT), and data analytics, the proposed model aims to improve the security and resilience of the food supply chain. The integration of advanced technologies and best practices will not only benefit the Omani food industry, but also provide valuable insights for global food traceability efforts.

Research gap

Context-specific models, the integration of cutting-edge technologies, a comprehensive approach to sustainability, stakeholder engagement, impact assessments of traceability systems, regulatory alignment, policy development, and consumer perceptions and trust are among the research gaps in Oman's food traceability industry [3]. While global research on food traceability is growing, there is a paucity of context-specific studies in Oman. Most frameworks and models have been developed in Western or high-income countries, with limited applicability to Oman's unique supply chain dynamics, cultural practices, and regulatory environments. The

current models are not customised to Oman's particular circumstances, and cutting-edge technologies like blockchain, IoT, and AI are not properly incorporated in line with Oman's perspectives [31]. The Sultanate of Oman also requires an all-encompassing strategy that takes the social, economic, and environmental aspects into account [3, 31]. Others discuss the extent of consumer centricity in the FSC adopting traceability systems [24, 35]. These studies have fragmented different traceability concepts and failed to identify a comprehensive and holistic model integrating elements such as technology, regulation compliance, and their impacts on food safety performance and sustainability practices.

Addressing these research gaps is essential for developing a comprehensive and sustainable food traceability model that enhances safety and resilience in the Oman food supply chain. Focusing on context-specific solutions, integrating advanced technologies, adopting a holistic approach to sustainability, fostering stakeholder collaboration, providing empirical impact assessments, aligning regulations, and understanding consumer perceptions are critical to optimising the food supply chains. Existing research often examines food traceability, sustainability, or resilience in isolation. There is a clear gap in comprehensive models that integrate all three dimensions to enhance overall supply chain performance, particularly in regions prone to environmental, logistical, and regulatory challenges like Oman. This study aims to fill the critical gaps and contribute significantly to both academic knowledge and practical applications in the field of food traceability. According to the existing literature, SC collaboration has great impacts on mediating mechanisms like data accuracy and timeliness, which eventually affect traceability and resilience [24, 56]. There is limited empirical evidence especially in the Middle East linking traceability systems to *tangible outcomes like improved consumer trust, food safety compliance, and crisis resilience*. This study offers a *data-driven validation* of these relationships.

This study gives a more collaborative explanation of traceability systems by embedding these elements into sustainable food traceability. It also discusses FSC traceability by examining the concept from a more comprehensive viewpoint using SC collaboration, regulatory framework compliance levels, traceability systems, and data timeliness and accuracy, which previous studies have fragmented.

Significance of the study

Creating a traceability model particularly tailored to the unique difficulties and circumstances of the Omani food industry fills a significant vacuum in the literature. This paradigm improves supply chain resilience and ensures

that food supplies remain stable and secure even during disruptions. It also improves food safety by facilitating the immediate detection and treatment of contamination and foodborne illness problems. This study informs FSC actors in Oman on the benefits of having an optimised FSC traceability system being influenced by SC collaboration, adherence to regulatory frameworks, and having accurate data, which eventually affect sustainable and food safety performance levels [34]. The traceability system has taken a major step towards modernity and efficiency with the integration of sophisticated technologies like blockchain, IoT, and AI. This integration provides a plan for utilising these technologies to enhance traceability in supply chains. In line with global sustainability goals and bolstering the food supply chain's long-term viability, the study promotes sustainable practices that are resource and environmentally friendly.

This research facilitates market access and trade for Omani food products by aligning local practices with international food safety standards, which is beneficial from a regulatory perspective. In addition to strengthening consumer trust, the model will focus on transparency and traceability and meet the growing need for information about the origin and food handling. Ultimately, this study provides recommendations for researchers, industry stakeholders, and legislators on how to improve food traceability and promote a safer, resilient, and sustainable food sector in Oman.

Theoretical framework

The study integrates numerous theoretical perspectives when discussing the dynamics of the global FSC and the impact of food safety, sustainable practices, and supply chain complexities on compliance standards, collaboration efforts, and data accuracy and timeliness. The information systems theory (IST), tasked with handling information transfer [23], is used in this study. The IST helps discuss the impact of traceability technologies on sustainability and food safety. Additionally, it helps discuss the dynamics of integrating traceability technologies, as well as challenges such as data accessibility, interoperability, and adoption rates that SC companies must overcome to achieve sustainability and food safety [3, 33, 34]. Similarly, the study uses institutional theory, which helps explain the relationship between organisational structures, norms, social relationship patterns, practices, and principles. The institutional theory provides a framework for understanding how institutions shape organisational behaviour, decision-making processes, and social change. The authors adopt this theory to help examine the relationships between stringent compliance levels, compliance verification efforts, and regulatory training programmes [17]. The theoretical

framework also highlights the benefits that regulatory compliance has on organisational behaviour and performance design.

The study also uses the data quality framework as a mediator. The framework is critical for analysing data accuracy and timeliness and the impact they have on technology traceability systems, food safety performance, and sustainable efforts [58]. The theoretical framework is crucial because it provides guidelines, methods, and tools that companies use to manage, improve, and ensure their data quality. This framework requires that all work be recorded and timestamped, which promotes organisation and timeliness. Additionally, it informs data verification rates and consistency by validating, cleaning, transforming, and monitoring data to achieve accuracy, consistency, completeness, reliability, and timeliness during use [29]. Further, the study utilises the economic theory of transaction cost on the influence of competitive self-interest [35]. This theoretical framework informs decisions by helping to predict the key forms of governance hierarchies, markets, or hybrids needed to organise economic activities through shared decision-making, information sharing, mutual trust, and communication effectiveness across the global FSC in its attempt to enhance resilience and sustainability through traceability systems.

Further studies reinforce the idea that the theory of total quality management (TQM), based on customer focus, continuous improvement, and teamwork, is crucial in understanding the relationship between food safety performance, incident rate, compliance with safety standards, the effectiveness of recalls, and the level of customer satisfaction to investigate the impact on the elimination of traceability problems [56]. An inclusive theoretical framework is used in this study, and it includes IST, institutional theory, data quality framework, and transaction cost theory, providing a basis for examining FSC security and traceability systems in Oman. This integrated approach provides an understanding of the interplay between technology traceability systems, regulatory compliance, organisational behaviour, and data quality. It also provides insights into the challenges and opportunities faced by SCs in achieving sustainability, food safety, and resilience in Oman.

Objectives of the study

1. To establish how the integration of traceability systems in the food supply chain affects sustainable practices in the Oman food industry.
2. To investigate how the implementation of traceability systems influences food safety performance in the food supply chain in the Oman food industry.

3. To assess the relationship between adherence to food safety standards and food safety performance in the Oman food industry.
4. To establish how adherence to standards influences sustainable practices in the Oman food industry.
5. To explore the mediating effect of data accuracy and timeliness on the relationship between technology-based traceability systems and food safety performance in the Oman food industry.
6. To establish the mediating role of collaborative supply chains in food safety performance in the Oman food industry.
7. To establish the mediating role of data accuracy and timeliness in the relationship between technology traceability systems and sustainable practices in the Oman food industry.
8. To evaluate the mediating role of collaborative supply chain relationships in the relationship between regulatory compliance and food safety performance in the Oman food industry.

Research questions

1. How does the integration of traceability systems in the food supply chain affect sustainable practices in the Oman food industry?
2. How does the implementation of traceability systems influence food safety performance in the food supply chain in the Oman food industry?
3. What is the relationship between adherence to food safety standards and food safety performance in the Oman food industry?
4. How does the adherence to standards influence sustainable practices in the food industry in the Oman food industry?
5. What are the mediating effects of collaborative supply chains on technology traceability systems and sustainable practice in the Oman food industry?
6. What are the mediating roles of collaborative supply chains in food safety performance in the Oman food industry?
7. How do data accuracy and timeliness mediate regulatory compliance and sustainable practices in the Oman food industry?
8. What is the mediating role of data accuracy and timeliness on regulatory compliance and food safety performance in the Oman food industry?

Hypothesis This study formulated the following hypotheses.

H₁: Traceability systems are associated with improved food safety performance in the Oman food industry.

H₂: Integration traceability systems are associated with improved sustainable practice in the Oman food industry.

H₃: Strong adherence to standards is positively associated with food safety performance in the Oman food industry.

H₄: Strong adherence to standards is positively associated with sustainability practices in the Oman food industry.

H₅: Collaborative supply chain mediates technology traceability systems and sustainable practice in the Oman food Industry.

H₆: Collaborative supply chain mediates technology traceability systems and food safety performance in the Oman food Industry.

H₇: Data accuracy mediates regulatory compliance and sustainable practice in the Oman food industry.

H₈: Data accuracy mediates regulatory compliance and food safety performance in the Oman food industry.

Literature review

Technology traceability systems, food supply chain safety, and sustainability

Traceability systems in the food supply chain refer to the ability to track and trace food products through all stages of production, processing, and distribution [18]. These systems ensure that detailed information about the origin, handling, and distribution of food products is recorded and accessible. The integration of advanced technologies has significantly enhanced the capabilities of traceability systems. The World Economic Forum states that food traceability systems (FTS) include the systematic arrangement of data collection, data storage, and data exchange processes that maintain and communicate desired product information across the different FSC levels [24]. The complexity of the global FSC presents numerous challenges to achieving shared product information symmetry, trust, and transparency [39]. Conversely, the fragmentation of the global FSC makes synergies and operational monitoring difficult [18, 26]. The maturity level of traceability technologies is the main barrier to smart adoption, diffusion, and integration rate [18]. FTSs should integrate and reconcile data from heterogeneous sources by quantifying the impact of their operations on food products, which ultimately helps make data-driven FSC decisions. Furthermore, they should be able to manipulate heterogeneous traceability records collected during the FSC process to support the calculation of safety, cost, and environmental sustainability [3]. In addition, it is believed that the architecture of distributed traceability systems has proven to be more resilient, risk-free, and sustainable

as it decentralises information collection to minimise the likelihood of traceability collapse and storage, which ultimately improves information symmetry, security, trust, and transparency [24]. On the other hand, the use of computer-assisted FTS has also been suggested as they are effective in identifying foods along the global FSC [54, 55]. For example, barcode technology is quickly being replaced by radio frequency identification (RFID) due to its huge data capacity, reading speed, and accurate data output. In addition, the FSC can use wireless sensors that support traceability-based product quality monitoring, electronic product code information services (EPCIS), blockchains, and smart contract technologies [5]. These technological FTSs have improved product data efficiency transfer between SC stakeholders and partners. Unfortunately, despite the revolutionary nature of FT technologies, their implementation and acceptance rates are still relatively low [49]. This is due to a lack of awareness and training on traceability and traceability technologies in the global FSC. Likewise, there are fewer standards in development, inefficient data management practices, resource limitations, and technology scaling issues that prevent SC companies from leveraging these technologies. Furthermore, FTSs have enabled stakeholders in SC to adopt more sustainable practices as they help a company gain a competitive advantage [15, 49]. A study adds that FTs help companies verify that good animal welfare practices are being followed [29]. They also help authenticate the legality of food production processes in the fishing industry and other unregulated sectors.

Food safety performance, sustainability, and regulatory compliance

Different regions and governments have regulatory framework standards designed to maintain food safety performance according to their expectations. It has been argued that the ability to integrate regulatory resources as intermediaries has a significant positive association with food safety and strengthens public crisis prevention and control [12]. However, due to the proliferation of heterogeneous data sources, traceability systems often face numerous interoperability problems and data integration problems [15]. They believe that FSC traceability systems still face these challenges as there is no universally accepted standard for how FSC companies record and transmit traceability data, which hinders the adoption of communication technologies that help maintain food safety performance [26]. Therefore, introducing and improving regulations related to food production, transportation, and operations is effective in reducing food safety violations, increasing compliance rates, increasing incident response time, strengthening social

management, and preventing crime in the field to reduce food safety [29].

There are food law requirements in several countries and regions, such as EU Regulation 178/2002, the Japanese Food Traceability Law, and the US Food Safety Modernization Act (FSMA). [24]. These ensure regulatory compliance from farm to market to improve food safety performance with SCs. Compliance with regulatory standards and laws is positively associated with sustainable practices [5]. For example, these regulatory standards are critical to reducing foodborne illness and minimising food waste and environmental degradation, leading to more sustainable and resilient food systems. The regulatory standards and laws surrounding green purchasing and sustainable packaging have positive impacts on SC's sustainability performance and resilience [9, 37].

Data accuracy and timeliness

Data accuracy and timeliness are critical to improving the safety performance and sustainability of the food supply chain. The use of food safety technology enables decentralised and tamper-proof, secure, and reliable FTS [62]. This leads to better traceability of the FSC as implementing reliable FTS technology resolves the problems of data inconsistency and verification [11]. On the other hand, FTSs should also examine the fragmented connections within the FSC to compile more information and thus achieve greater sustainability [27]. Food safety supply chains collect large amounts of data that require accurate sorting, classification, and verification to achieve consistency [46]. Additionally, more accurate and timely on-chain data can reduce callback response times. This interaction highlights the mediating role of data accuracy and timeliness in maintaining food safety performance levels [19]. Tajima [57] takes an alternative viewpoint and finds that the level of data accuracy improves sustainability metrics. Ideally, quality and efficient data timeliness promote accuracy and data reliability for hazard analysis, sustainable practices, and preventive controls.

The reliability and security of the supply chain are significantly increased by the timely and accurate data. All parties involved in the food supply chain are guaranteed access to the fast and accurate data they need to make informed decisions, address problems promptly, and maintain the integrity of the chain. Hazard detection, regulatory compliance, consumer trust, and operational efficiency are improved with timely and accurate data. The combination of cutting-edge technologies such as blockchain, IoT, and AI improves the timeliness and accuracy of data, enabling rapid problem resolution and real-time insights. The food company can create a more robust and reliable supply chain that ensure consumer safety and satisfaction while complying with regulatory

requirements by placing high priority on data timeliness and accuracy.

Impact of collaborative supply chain relationships on regulatory compliance, food safety performance, and sustainability

Supply chain collaboration (SCC) enables actors and organisations to achieve SC outcomes together. It was added that SCC is crucial in modern SC practices as it can lead to better performance levels by redesigning the workflow and promoting resource sharing between SC members and actors. The United Nations Sustainable Development Goals (UN SDGs) emphasise cross-sectoral and cross-country cooperation and cooperation in achieving sustainable goals [49]. Collaboration between stakeholders has a positive association with improved regulatory compliance and food safety performance. When FSC stakeholders and actors work together to improve traceability and incident reporting by promoting effective communication, making shared decisions, and building mutual trust, they make the global FSC more resilient [12]. A more collaborative FSC contributes to achieving environmental benefits and SDGs as implementing environmentally friendly practices is easier when SCC is adopted [46].

Supply chain collaboration becomes more effective when actors share real-time information and leverage high-level supply chain redesign to achieve goals such as improved traceability [10]. SCC makes the SC more comprehensive as it informs stakeholders and actors to work together to reduce carbon footprints and waste and increase community engagement [44]. Furthermore, the level of regulatory compliance increases when FSC actors and stakeholders work together to find a common way to improve food safety performance levels [10]. Collaboration across the entire supply chain has a major impact on sustainability, food safety, and regulatory compliance. These partnerships improve the overall resilience and effectiveness of the food supply chain by promoting openness, standardisation, and collaborative efforts. In the context of the food industry, fostering collaborative relationships between stakeholders can lead to a safer, more compliant, and sustainable supply chain, benefiting all stakeholders and supporting the long-term sustainability of the sector.

Methodology

This study adopts a positivist research philosophy focused on quantitative research designs and approaches [52]. Positivist philosophy is valued for its in-depth insights and analysis of research using multiple variables with hypothesis testing and conceptual models. Studies also state that positivist philosophy is ideal for testing

causality between variables based on the researcher's theory and assumptions [22]. Partial least squares structural equation modelling (PLS-SEM) was used to analyse the results of this study. The procedure includes the analysis of the following aspects: structural modelling and model specification, data acquisition and PLS path model estimation, creation of the measurement model, and evaluation of the structural model. Data were collected using online questionnaires via Google Forms.

Questionnaire development and administration

Survey-based studies often use questionnaires as a data collection tool. This study utilised existing questionnaire validation [63] to ensure high-quality responses and results. The variables and indicators were adopted from the existing literature on sustainable food traceability. A sample of 385 participants was selected from the Oman Logistics Association (OLA), logistics scholars at universities in Oman, and food industry executives. Participants were selected based on a previous study [28], whose predetermined sample size calculations represented a population of 1 million with a 95% confidence level and a 5% margin of error. The questionnaire design involved identifying key constructs aligned with the research objectives, such as technological traceability, collaborative supply chain relationships, data accuracy and timelines and regulatory compliance. These constructs were operationalised into measurable variables using multi-item scales like Likert scales and binary responses. Validation was conducted through expert reviews for content validity, pilot testing for clarity, and statistical analyses like factor analysis and reliability testing (Cronbach's Alpha). Adjustments based on feedback ensured the questionnaire captured the constructs accurately and was user-friendly. This process ensured the instrument was reliable, valid, and capable of producing actionable insights. The questionnaire was pilot tested on academics from the Middle East College and the University of Technology and Applied Sciences in Oman. Participants were selected using a snowball technique that allowed them to share the link to the questionnaire on platforms such as LinkedIn, WhatsApp, and email from 1st November, 2023, to 15th January, 2024. Snowball sampling presents challenges such as *sampling bias*, as the process often begins with a small group of initial participants whose characteristics may not represent the broader target population [20]. This bias can become compounded as participants refer others within their social or professional circles, creating a homogenous sample. To address this, researchers diversified the initial participants and used multiple entry points to access different subgroups within the population. Another challenge is the *over-reliance on social networks*, which can limit access to

participants outside the existing networks of the initial group, excluding less-connected or marginalised individuals [43]. Therefore, the researchers actively sought participants from diverse backgrounds or underrepresented communities which included other nationalities and professionals from other institutions and industries in Oman. Additionally, Mohd Ishak and Abu Bakar [40] highlight *data quality concerns* may arise as some referred participants may not fully meet the inclusion criteria or may provide incomplete data. To counter this, researchers established clear inclusion criteria and verified participants' eligibility at each stage. *Ethical considerations* of maintaining confidentiality in a chain-referral process can be challenging, requiring strict protocols to protect participants' privacy and ensure informed consent throughout the process [41]. Overall, 60% of the participants were Omanis, 10% were Pakistanis, 20% were Indians, 5% were Bangladeshis, and 5% were from other countries. The diversity of nationalities involved was important to obtain different views on the study. In terms of gender diversity, 70% were male and 30% were female. The participants consisted of 10% lower-level managers, 50% middle-level managers, 15% top-level managers, and 10% were owners of the companies. Diversity in terms of management was important as the results encompass all levels of management important for sustainable traceability in the food industry. Most participants had higher qualifications in SC and had worked in the food industry for more than 5 years. Participation in the study was voluntary, and ethical approval was obtained from the Middle East College Ethics Review Committee (MECERC) (number MEC00426) before administering the questionnaire. All ethical considerations were listed on the cover sheet of the questionnaire to allow participants to fully understand the nature of the study. The cover page of the questionnaire highlighted the use of informed consent to obtain participants' explicit consent to participate.

Indicator and construct explanations

The specification of the structural model was based on a literature review and research assumptions as well as an understanding of the field. This study consists of traceability technology systems (TTS) and regulatory compliance (RC) mediated by collaborative supply chain relationships (CSCRs) and data accuracy and schedules (DAT) as indicated in Fig. 1. The dependent variable is sustainable practices (SP) and food safety performance (FSP). Traceability technology systems consist of three indicators: traceability coverage (TTS1), error rate and data accessibility (TT2), and integration with other systems (TT3). Regulatory compliance is measured using four indicators: compliance

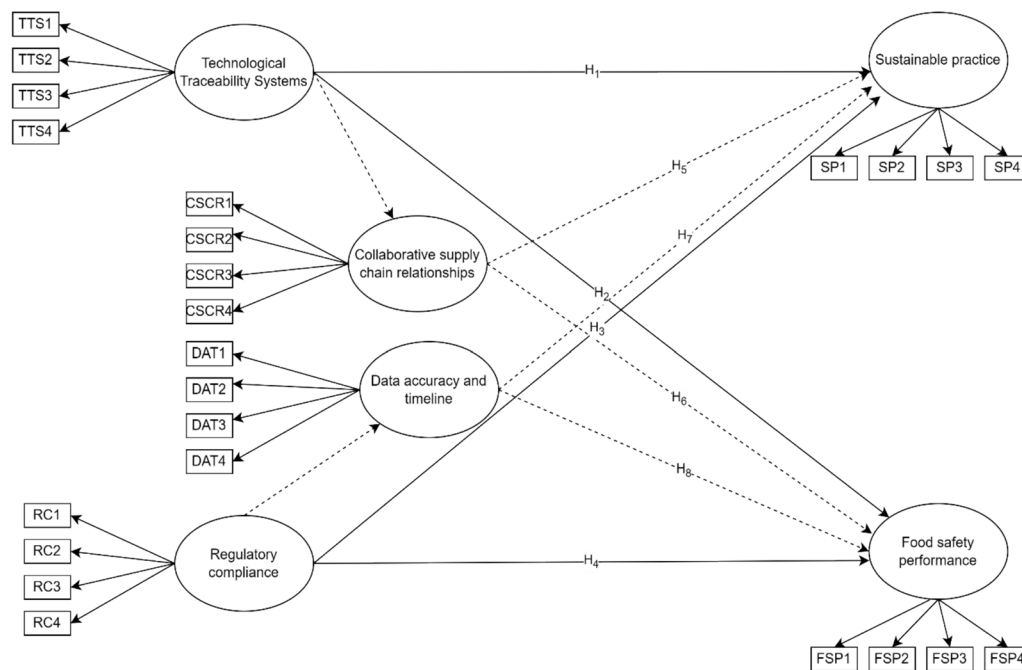


Fig. 1 Conceptual model. Source: Primary data

(RC1), audit success rate (RC2), employee training (RC3), and number of compliance violations (RC4). The mediating variable indicators of supply chain collaboration are supplier collaboration index (CSCR1), information sharing and effectiveness (CSCR2), joint product development project (CSCR3), and risk reduction (CSCR4). The dependent variables for sustainable practice indicators are carbon footprint reduction strategies (SP1) and employee training (CSCR4), while food safety performance indicators are customer complaint reduction (FSP1), audit performance ratings (FSP2), participation in training programmes (FSP3), and Audit are performance (FSP4). The framework illustrates the interconnected relationships between key elements, demonstrating how one construct influences another to varying degrees. It highlights how effective systems can foster collaboration within supply chains, emphasising their role in enhancing relationships. Collaboration, in turn, plays a critical role in driving performance outcomes, underscoring its importance in achieving standards and improving efficiency. Additionally, the framework identifies areas where contributions, while positive, are relatively weaker, suggesting the need for complementary strategies or interventions to enhance their impact. Overall, the framework provides a comprehensive view of the relationships, guiding the development of targeted strategies to optimise outcomes and address areas of limited influence.

Results and analysis

In conformance with the PLS-SEM procedure, the study used SMART PLS 4 to analyse the data due to its ability to measure hypotheses reflecting relationships between constructs and indicators [21]. Using SMART PLS 4 for PLS-SEM analysis offers significant benefits for the study of sustainable food traceability in Oman. The software's ability to handle complex models, its user-friendly interface, robust evaluation metrics, and flexibility make it an ideal choice for developing and testing a comprehensive model aimed at enhancing safety and resilience in the food supply chain. Leveraging on the capabilities of SMART PLS 4, the study can achieve a thorough and insightful analysis, ultimately contributing to better regulatory compliance, improved food safety performance, and greater sustainability in the Oman food industry [14]. The study assessed the formative measurement using reliability and validity assessments. This was followed by an assessment of the structural model to confirm the study hypothesis. The next section reflects on the construct reliability and validity of the study.

Reliability and validity

Table 1 presents the study construct reliability and validity and factor loading of the study. All values in the study needed to be above the threshold of 0.7 to confirm construct reliability and validity as recommended by previous research [13]. The study indicators were all above the

Table 1 Factor loading, Cronbach's alpha AVE, CR, Rho_A, and CA

	Factor loading	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Collaborative supply chain relationships	0.879 0.808 0.790 0.862	0.860	0.900	0.902	0.699
Data accuracy and timeliness	0.910 0.928 0.912 0.905	0.934	0.937	0.953	0.835
Food safety performance	0.825 0.842 0.850 0.859	0.866	0.867	0.908	0.712
Regulatory compliance	0.886 0.933 0.917 0.859	0.921	0.921	0.944	0.809
Sustainable practice	0.938 0.908	0.829	0.850	0.920	0.853
Technology traceability systems	0.924 0.914 0.815	0.861	0.869	0.916	0.784

0.7 benchmark, reflecting a good measure of the latent variables as recommended by an existing study [25]. These results imply that the constructs have satisfactory internal consistency and reliability and can be used in assessing the model. The study proceeded to carry out the discriminant validity using the heterotrait-monotrait ratio, (HTMT) and Fornell and Larcker Criterion as recommended by a prior research hair [60].

Discriminant validity

In comparing the correlation between constructs, it is recommended [38] that discriminant validity using the HMTM ratio should be below 0.85, which all variables in

Table 2 were. This suggests that the constructs in Table 2 are distinct, and the measurement model is sound. This further strengthens the credibility of the results, which enhances the confidence in the relationships established between the latent variables. Discriminant validity was confirmed by the Fornell and Larcker criterion in Table 3. Although HMTM and the Fornell and Larcker criterion measure the discriminant validity, it is recommended to use both methods as a way of increasing the robustness and credibility of the measurement model [2]. The Fornell and Larcker criterion, unlike HTMT, compares the square root of AVE for each construct with the correlation between that construct and other constructs [13].

Table 2 Heterotrait-monotrait ratio (HTMT)

	Collaborative supply chain relationships	Data accuracy and timeliness	Food safety performance	Regulatory compliance	Sustainable practice	Technology traceability systems
Collaborative supply chain relationships						
Data accuracy and timeliness	0.165					
Food safety performance	0.454	0.673				
Regulatory compliance	0.599	0.338	0.627			
Sustainable practice	0.635	0.159	0.501	0.658		
Technology traceability systems	0.506	0.200	0.416	0.652	0.412	

Table 3 Fornell and Larcker

Collaborative Supply Chain Relationships	0.836					
Data Accuracy and Timeliness	0.133	0.914				
Food Safety Performance	0.425	0.610	0.844			
Regulatory Compliance	0.558	0.315	0.562	0.899		
Sustainable Practice	0.553	0.144	0.431	0.578	0.923	
Technology Traceability Systems	0.458	0.169	0.364	0.584	0.352	0.886

The AVE is a measure used in SEM to assess the discriminant validity of constructs. Discriminant validity is then confirmed if the AVE for a construct is greater than the correlations with other constructs. In this study, all the square roots of the AVE are greater than the correlations with other constructs, hence discriminant validity is achieved. This implies that the constructs are distinct from each other and can be used in the measurement model [22].

Direct relationships

After confirming the construct validity, reliability, factor loading, and discriminant validity, the study went on to

test the direct and indirect relationships of the model as indicated in Tables 4 and 5.

Pathway Diagram Key

- TTS: Technology Traceability Systems
- CSCR: Collaborative Supply Chain Relationships
- DAT: Data Accuracy and Timeliness
- RC: Regulatory Compliance
- SP: Sustainable Practices
- FSP: Food Safety Performance

The values between constructs (e.g. 0.210, 0.099, and 0.550) are path coefficients, indicating the strength and direction of relationships between latent constructs, as shown in Fig. 2. The direct effects in the model illustrate the varying degrees of influence among constructs

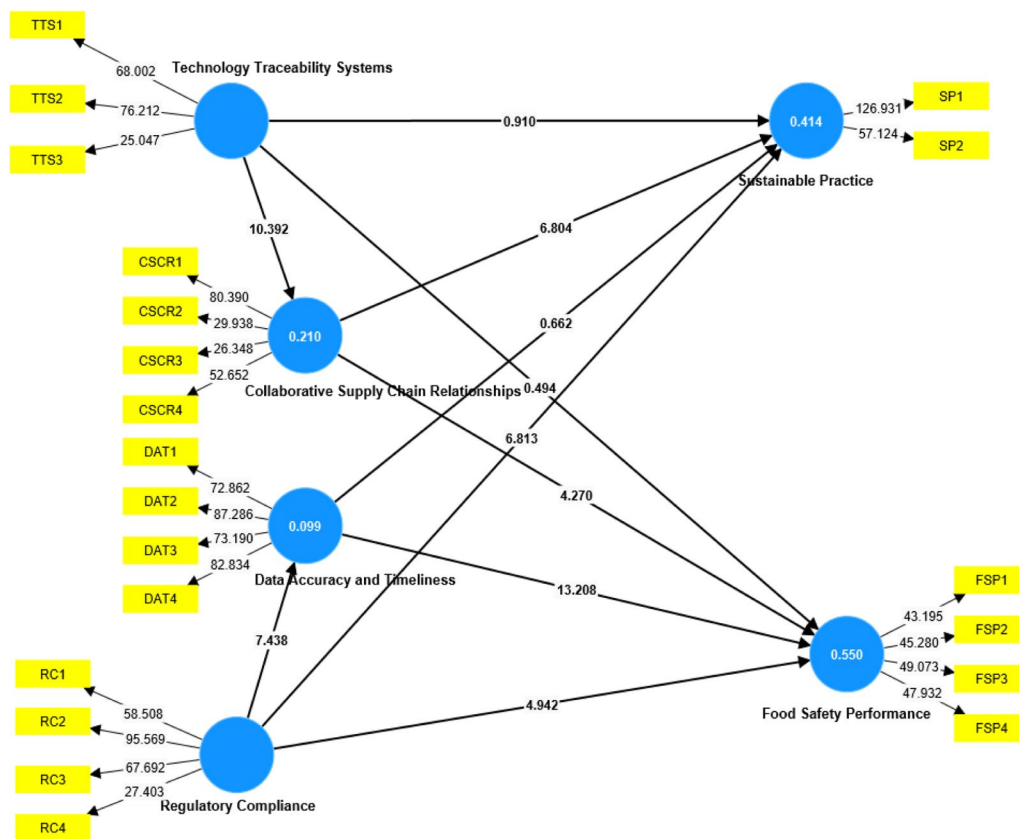
**Fig. 2** PLS predict model pathway diagram

Table 4 Direct relationships

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T-statistics (O/STDEV)	P values
H ₁ : Technology Traceability Systems—> Sustainable Practice	0.108	0.113	0.053	2.044	0.041
H ₂ : Technology Traceability Systems—> Food Safety Performance	0.113	0.117	0.056	2.006	0.045
H ₃ : Regulatory Compliance—> Food Safety Performance	0.442	0.437	0.056	7.939	0.000
H ₄ : Regulatory Compliance—> Sustainable Practice	0.416	0.412	0.059	6.998	0.000

Table 5 Indirect relationships

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T-statistics (O/STDEV)	P values
Regulatory Compliance—> Data Accuracy and Timeliness—> Food Safety Performance	0.154	0.154	0.022	6.970	0.000
Regulatory Compliance—> Data Accuracy and Timeliness—> Sustainable Practice	− 0.009	− 0.008	0.013	0.652	0.514
Technology Traceability Systems—> Collaborative Supply Chain Relationships—> Food Safety Performance	0.085	0.088	0.020	4.186	0.000
Technology Traceability Systems—> Collaborative Supply Chain Relationships—> Sustainable Practice	0.157	0.158	0.027	5.700	0.000

within the system. The relationship between technology traceability systems (TTS) and collaborative supply chain relationships (CSCRs) is moderately positive, with a path coefficient of 0.210, suggesting that effective traceability systems foster stronger collaboration within supply chains. Collaborative supply chain relationships (CSCRs) have a significant and strong direct effect on food safety performance (FSP), as indicated by the high path coefficient of 0.550, underscoring the critical role of collaboration in ensuring food safety standards. Meanwhile, the relationship between data accuracy and timeliness (DAT) and sustainable practices (SP) is weakly positive, with a path coefficient of 0.099, indicating that while accurate and timely data contribute to sustainable practices, the influence is limited and may require additional factors to achieve a stronger impact. These findings highlight the varied contributions of different constructs and emphasise the need for targeted strategies to optimise their effects.

Factor loadings represent the degree to which each observed variable contributes to its associated latent construct, as shown by the numbers next to the yellow rectangles. These values highlight the strength of the relationship, with higher loadings (closer to 1) indicating a stronger and more reliable contribution. For example, in the case of technology traceability systems (TTS), the observed variables TTS1 (68.002), TTS2 (76.212), and TTS3 (25.047), all contribute to the latent construct TTS. Among these, TTS2 demonstrates the strongest

contribution, indicating its greater relevance and reliability in defining the construct compared to the other observed variables.

Discussion of results

The results section delves into the analysis of both direct and indirect relationships among the constructs within the proposed model. This analysis highlights the strength and direction of key pathways, offering insights into how various factors interact to influence safety, sustainability, and resilience in the food supply chain. These relationships provide a comprehensive understanding of the dynamics at play, setting the foundation for informed discussions and practical recommendations. The findings are presented systematically to illustrate their significance and implications for the Oman food industry.

Path analysis: direct relationships

Direct PLS-SEM relationships refer to links between two constructs with a single arrow [2]. In this study, the direct relationships are between independent variables (IVs) and dependent variables (DVs). The results are presented and discussed in the next section.

H₁: Integration traceability systems are associated with improved sustainable practice in the Oman food industry.

Technology traceability systems in Table 4 indicate a positive impact on technology traceability systems indicate a positive impact on collaborative supply chain relationships. This precision promotes confidence among

supply chain partners, leading to smoother collaboration as participants can rely on the data provided by the traceability system. The findings are supported by existing research that emphasises that industrial traceability systems function within complex SCs with a diverse array of participants [8]. Thus, business operators are tasked with ensuring the fulfilment of relevant information requirements and verifying compliance within the operations under their purview. As indicated by [24], the adoption of a traceability system in the supply is fundamental to making the supply chain more resilient, risk-free, and sustainable. This comes from the effective decentralisation of information which eventually improves information symmetry, security, trust, and transparency [24]. On the other hand, the use of computerised FTSs has also been suggested because they are effective in identifying food along the global FSC [54, 55].

H₂: Traceability systems are associated with improved food safety performance in the Oman food industry.

Results indicate that technology traceability systems have a positive impact on food safety performance. A technological traceability system allows for quick identification and isolation of potentially unsafe products. This can lead to a reduction in customer complaints related to issues like contamination and inaccurate labelling. Putting the systems in place that allow customers to make complaints may signify improved food safety. It indicates that the traceability system is effective in preventing and addressing issues before they reach the end consumer. A similar study highlighted that traceability and food safety are linked [29]. Therefore, the results imply that technology traceability systems play a significant role in collaboration and guaranteeing food safety performance. These results are important in understanding the importance of traceability systems and technologies in Oman. The same assertion is supported by [3], who assert that the incorporation of traceability systems in the food industry allows the integration and alignment of heterogeneous data that may be affecting operations and supply chains in the food industry. This will foster stakeholders to make data-driven decisions that are important in calculating safety, cost, and food safety performance.

H₃: Strong adherence to standards is positively associated with food safety performance in the Oman food industry.

Results indicate that regulatory compliance has a strong positive impact on food safety. A high audit success rate indicates that the food supply chain entities can meet and exceed required regulatory standards. This success rate is reflective of a commitment to best practices and adherence to safety protocols. A positive audit outcome will lead to improved food safety performance. Similarly, a previous study highlighted HACCP

as a regulatory innovation to enhance food safety within the meat industry [7]. The results presented in Table 4 confirm the need for Oman to improve its adherence to standards to enhance food safety performance. This is consistent with the assertion by [3] that the establishment and continuous improvement of food regulations in production, transportation, and operations is an effective way to eliminate food safety violations and reduce food safety crimes. Relatedly, training of stakeholders on food safety regulations and compliance remains a relevant factor to compliance aspects as stated in countries such as EU Regulation 178/2002, Japan's Food Traceability Act, and the US Food Safety Modernization Act (FSMA) [24] that have regulatory compliance from farm to market to improve food safety performance with SCs.

H₄: Strong adherence to standards is positively associated with sustainability practices in the Oman food industry.

The findings indicated that regulatory compliance has a strong positive impact on sustainable practices. A lower number of compliance violations indicate a commitment to maintaining high standards. Minimising violations suggest effective management of environmental impacts and sustainable initiatives. This result corroborates the study that indicated that controls, legitimacy, and deterrence are important to the sustainability practices of manufacturing companies in Uganda [53]. This observation shows that regulatory compliance by government and food industry players plays an important role in the FSC. Research by [9, 37] has also revealed that adherence to the regulatory standard is a critical component in reducing foodborne illness, and minimising waste leading to sustainable and resilient food systems. Critical to the finding is that Oman's legislative standards and laws that promote environmental purchasing and packaging have positive outcomes when it comes to SC sustainability performance and resilience.

Path analysis using bootstrapping: indirect relationships

Indirect relationships are also known as mediating relationships. They have been regarded as relationships that exist with at least one intervening variable involved [16]. In this study, the intervening variables of IV and DV are collaborative supply chain relationship and data accuracy and timeliness. In Table 5, the results show that all the t-statistics are significant at ($t < 0.05$). This suggests the presumed effects are not due to chance [51].

H₅: Collaborative supply chain mediates technology traceability systems and sustainable practice in the Oman food Industry.

Technology traceability systems → collaborative supply chain → sustainable practice, and thus (H_5) is supported. *Original Sample (O): 0.154*: The indirect effect is positive,

indicating that improved regulatory compliance enhances data accuracy and timeliness, which in turn improves food safety performance. The *T-Statistics: 6.970, p-Value: 0.000*: The effect is statistically significant, meaning the relationship is unlikely due to chance. Technology traceability systems can enhance visibility in the supply chain by providing stakeholders with real-time data on the movement and status of products. This transparency helps in identifying inefficiencies, reducing errors, and ensuring that sustainable practices are followed. Thus, related to food safety, the results suggest that stronger supply chain collaboration enabled by traceability systems leads to sustainable practice. This study implies that integrating technology traceability systems into collaborative FSCs in Oman has the potential to enhance monitoring, response capabilities, compliance, and consumer confidence. This can contribute to the overall sustainability of the FSC by reducing waste and improving the efficiency of processes. However, implementing such practices requires the commitment and cooperation of all stakeholders involved in the FSCs in Oman.

H_6 : Collaborative supply chain mediates technology traceability systems and food safety performance in the Oman food Industry.

Technology traceability systems $\rightarrow$ collaborative supply chain relationships $\rightarrow$ food safety performance, and thus (H_6) is supported. *Original Sample (O): 0.085*: The indirect effect is positive, showing that technology traceability systems enhance collaborative supply chain relationships, which subsequently improve food safety performance. *T-Statistics: 4.186, p-Value: 0.000*: This relationship is statistically significant, confirming its reliability. When organisations utilise advanced technology to trace their products in the SC and foster collaboration among supply chain partners, it positively impacts food safety performance. This could be due to increased transparency, better communication, and more efficient response mechanisms in food safety issues. The significance of the path indicates that the incorporation of traceability systems in the supply chain improves supply chain collaboration and thus leads to food safety performance.

H_7 : Data accuracy mediates regulatory compliance and sustainable practice in the Oman food industry.

Regulatory compliance $\rightarrow$ data accuracy and timeliness $\rightarrow$ sustainable practice (H_7), indicating that the hypothesis is not supported. *Original Sample (O): -0.009*: The indirect effect is negligible and negative, suggesting that regulatory compliance, through data accuracy and timeliness, and has little to no meaningful impact on sustainable practices. The *T-Statistics: 0.652, p-Value: 0.514*: This effect is not statistically significant, indicating no strong evidence of this indirect relationship. This result is not

surprising, but it reinforces the fact that strict regulatory compliance by governments and regulatory authorities does not result in sustainable compliance. This lack of support for the hypothesis suggests that a more holistic approach is required for the FSCs in Oman to achieve sustainable compliance with regulatory standards. This could involve a combination of regulatory measures, technological advancements, continuous monitoring, and collaborative efforts among stakeholders.

H_8 : Data accuracy mediates regulatory compliance and food safety performance in the Oman food industry.

Regulatory compliance $\rightarrow$ data accuracy and timeliness $\rightarrow$ food safety performance (H_8). *Original Sample (O): 0.085*: The indirect effect is positive, showing that technology traceability systems enhance collaborative supply chain relationships, which subsequently improve food safety performance. *T-Statistics: 4.186, p-Value: 0.000*: This relationship is statistically significant, confirming its reliability. The path is significant, indicating that if companies improve regulatory compliance, it will result in strengthening data accuracy and timeliness, hence resulting in food safety performance. By focusing on these aspects, food supply chains in Oman can enhance their overall resilience, trustworthiness, and commitment to delivering safe and high-quality food products to consumers.

Conclusion

In summary, compliance plays a critical role in improving food safety within food supply chains in Oman. Complying with regulations can improve the accuracy and timeliness of their data management processes, resulting in improved food safety. The improvement in food safety performance is attributed to the collaborative relationships within the SC facilitated by technology traceability systems. As a result, the implementation of technology traceability systems in Omani FSCs contributes to improving sustainable practices within the SC. This study also suggests that regulatory compliance and implementation of technology traceability systems play an essential role in improving sustainability and food safety in Omani FSCs. The model presented in this study provides a calculated method for changing the food supply chain in Oman. It uses cutting-edge technologies and international best practices to address the sector's particular difficulties. The implementation of this approach ensures that the Oman food industry improves its safety standards, adheres to strict rules, and promotes sustainable practices—which will ultimately contribute to consumer safety and the industry's longevity.

In indirect relationships in the Oman food industry, regulatory compliance and technology traceability systems positively impact food safety performance

through intermediary variables (data accuracy and collaborative relationships). Technology traceability systems also support sustainable practices via collaborative relationships. Regulatory compliance has no significant indirect effect on sustainable practices through data accuracy. To ensure the long-term sustainability and safety of Oman's food industry, policymakers must prioritise the development and enforcement of robust, technology-enabled traceability frameworks. This includes investing in infrastructure that supports the adoption of IoT, blockchain, and AI, as well as creating incentives for supply chain actors to adopt sustainable practices. By fostering collaboration among stakeholders, these findings highlight that while regulatory compliance strongly supports food safety, its link to sustainability is weaker. Conversely, technology traceability systems enhance both food safety and sustainability through supply chain collaboration.

Theoretical implications

Food security is an important component of a country's health status. The study makes significant contributions to multiple theoretical domains. Integrating concepts from supply chain management, food safety, sustainability, technology adoption, stakeholder theory, and consumer behaviour enables a holistic framework for improving safety and resilience in the Omani food industry while advancing theoretical understanding in the field. It supports the use of technology traceability, compliance measures, sustainable practices, and food safety performance. Integrating concepts from supply chain management, food safety, sustainability, technology adoption, stakeholder theory, and consumer behaviour enables a holistic framework for improving safety and resilience in the Omani food industry while advancing theoretical understanding in the field. It supports the use of technology traceability, compliance measures, sustainable practices, and food safety performance. The study demonstrates that technology traceability systems positively impact collaborative supply chain relationships and food safety performance. This finding aligns with and extends the existing research by reinforcing the importance of traceability systems in improving transparency, efficiency, and safety within food supply chains. Previous studies have highlighted similar benefits, but this research adds specific evidence from the Omani context. This emphasises the role of advanced technologies in promoting sustainable practices. The findings highlight the critical role of collaborative relationships within the supply chain in improving food safety and sustainability. Organisations can foster better collaboration, which in turn enhances supply chain transparency and efficiency. This supports existing theories on the importance of

collaboration in supply chain management and provides insights for implementing such systems in the food industry.

Practical implications

This study is aimed at management and captains of industry in the food industry in Oman and around the world. It provides insights into the best way companies can calibrate their food supply chains by integrating technology into product traceability. In line with the managerial implications, the study recommends investing in advanced traceability systems such as blockchain, IoT, and AI to improve food safety, regulatory compliance, and supply chain resilience. In addition, it recommends collaboration between supply chain partners, providing training and capacity-building programmes, adopting sustainable practices, and ensuring compliance with food safety and traceability regulations. A culture of continuous improvement and innovation and educating consumers about the benefits of traceability systems and sustainable practices are also recommendations from the study. These recommendations aim to improve operational efficiency, cost-effectiveness, and market competitiveness in the Omani food industry. Supply chain transparency is becoming increasingly important for building competitive advantage, and supply chain collaboration remains a core part of business performance. This study outlines the importance of supply chain collaboration and how it can be used to improve supply chain agility towards sustainable practices.

Regional implications

The study's regional implications for Oman's food industry are significant given the country's growing focus on food security, sustainability and supply chain resilience. As Oman continues to diversify its economy and reduce reliance on imports, implementing sustainable food traceability systems will become critical. This comprehensive model provides a framework for improving security and resilience by fostering collaboration among stakeholders, improving data accuracy, and integrating advanced technologies tailored to regional needs. Promoting transparency and efficiency within supply chains will address challenges specific to Oman, such as the impact of the dry climate on agriculture and dependence on imports. Furthermore, the model is in line with the objectives of Oman's Vision 2040 and supports sustainable practices and innovations in key sectors. By implementing these strategies, Oman can increase consumer confidence, ensure compliance with international safety standards and position itself as a leader in sustainable food supply chains in the Gulf Cooperation Council (GCC) region.

Limitations and future research directions

The study was conducted in the context of the Omani food industry, and it may be useful to include information from other industries that may use traceability technology systems in supply chains. Therefore, it may be difficult to generalise the results of the study. The study adopted a positive approach to data collection and analysis. However, more comprehensive insights could have been obtained if qualitative philosophies had been applied to the study. The mono-method of data collection and analysis using questionnaires and Smart PLS is rigid, and incorporating other analysis methods could lead to new insights into the study. The study used data collected over a short period, which would potentially limit its scope and results. Data collected over a longer period, covering a broader area of sustainable food safety, could provide much deeper knowledge. Based on the above limitations, further research can develop new approaches and insights in the study area.

Future research on technological advancements, cross-industry comparisons, consumer behaviour, policy and regulatory analysis, supply chain resilience, risk management, sustainability, and cross-sector collaboration will enhance knowledge about food traceability. Researchers can also explore the environmental impact of traceability technologies and suggest collaboration between sectors to address common challenges and promote innovation in the Oman food industry. Furthermore, researchers should investigate the role of indigenous knowledge systems and grassroots movements in advancing sustainable and sovereign food systems and how these systems can be integrated with modern technology. In effort to make research results applicable, research should focus on context-specific implementation strategies and evaluate the impact of traceability systems on consumer trust, regulatory compliance, and crisis resilience.

Novelty of the research

The study provides a unique understanding of the food supply chain management sector in the Sultanate of Oman. Robust data analysis provides unique insights into the impact of traceability technology systems on improving sustainable food safety practices. This reflects the role of regulatory compliance in improving food safety and offers a new perspective on policy developments in Oman, the Middle East and other economies. The systematic data collection and analysis using the mediation effect of the collaborative supply chain relationship and data accuracy deepen the understanding of the dynamics in Oman's food supply chain and make an insightful regional case study. The

novelty of this study also lies in the development of a comprehensive model tailored to the unique context of the food industry in Oman, taking into account both security and resilience within the supply chain. Unlike generic frameworks, this study integrates sustainable practices, advanced traceability systems and collaborative supply chain relationships specifically designed to address the challenges Oman faces, such as the heavy dependence on imports, climatic restrictions, and changing consumer demands. Incorporating local and regional dynamics into the model allows it to provide actionable insights to improve food security while promoting long-term sustainability and resilience. In addition, the study combines theoretical concepts with practical applications and offers a holistic approach. This targeted approach represents a ground-breaking effort to leverage sustainability, technology, and collaboration to improve the region's food supply chain.

Abbreviations

SC	Supply chains
SCM	Supply chain management
FSC	Food supply chain
FSCM	Food supply chain management
PLS-SEM	Partial least squares structural equation modelling
OLA	Oman Logistics Association

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Author contributions

All authors Dr RM, Ms JSA, Dr CM, and Mr CT wrote the manuscript, analysed the data, and proofread the final document.

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Availability of data and materials

The necessary data for the research findings may be obtained by submitting a written request to the grant funder, which in this case is the Ministry of Higher Education, Research, and Innovation in the Sultanate of Oman, as well as the Middle East College principal researcher.

Declarations

Ethics approval and consent to participate

The contributors confirm that the paper was not submitted for publication. Participation in the study was voluntary and ethical approval was obtained from the Middle East College Ethics Review Committee (MECERC) (number MEC00426) before administering the questionnaire. All ethical considerations were listed on the cover sheet of the questionnaire to allow participants to fully understand the nature of the study. The cover page of the questionnaire highlighted the use of informed consent, to obtain participants' explicit consent to participate.

Consent for publication

All authors agreed to publish the paper under Future Business Journal.

Competing interests

The authors declare that there is no conflict of interest.

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